LAPORAN PRAKTIKUM

POSTTEST (5)

ALGORITMA PEMROGRAMAN LANJUT

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PROGRAM STUDI INFORMATIKA

UNIVERSITAS MULAWARMAN

SAMARINDA

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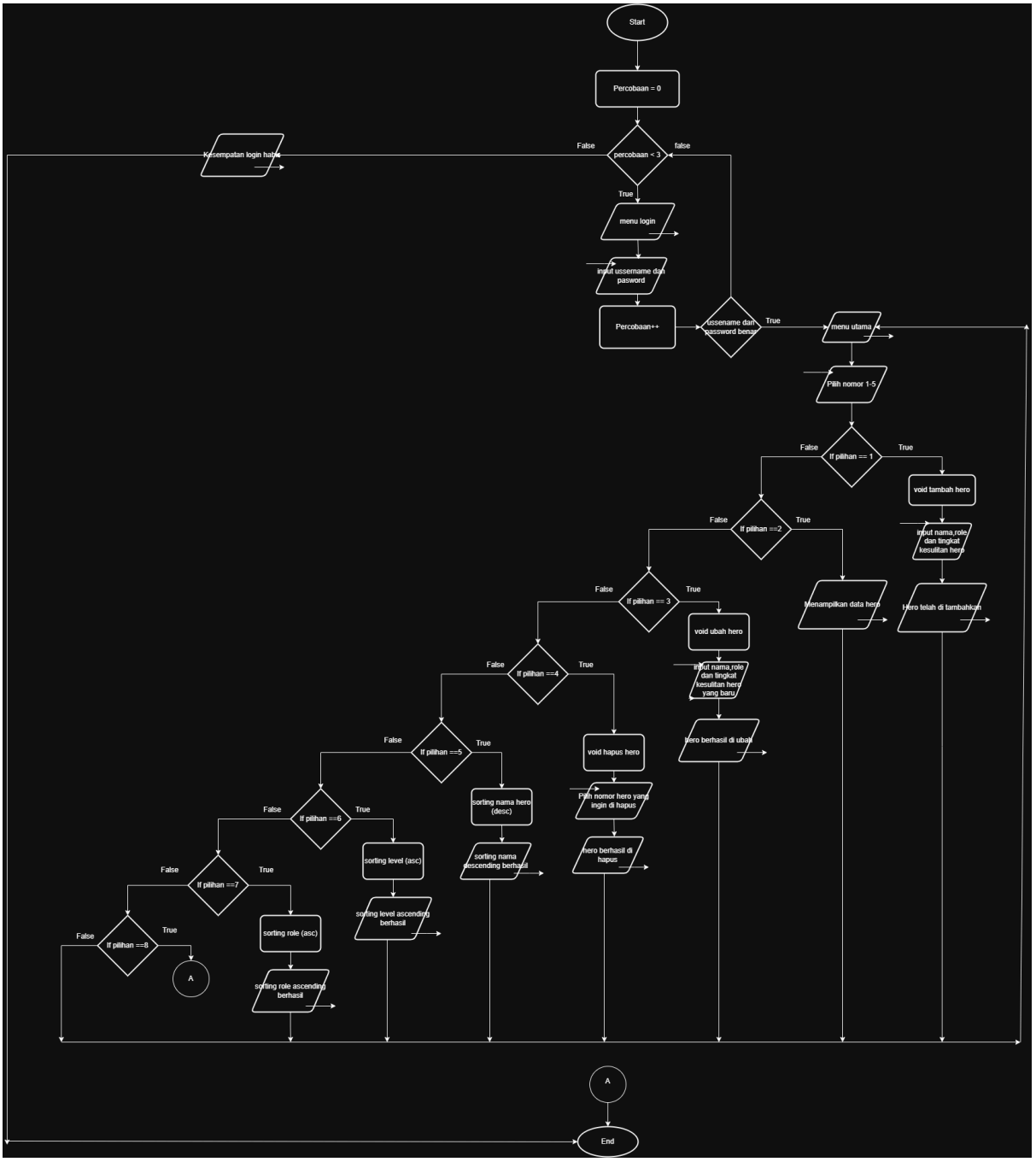
1. Flowchart

pertama dimulai dari proses login. Program dimulai dengan inisialisasi percobaan login dan pengguna diminta memasukkan username serta password. Jika login berhasil, pengguna akan masuk ke menu utama, sedangkan jika gagal percobaan akan bertambah hingga maksimal tiga kali, dan jika sudah habis program akan berakhir. Setelah berhasil login, pengguna dapat memilih menu 1–8 yang berisi fitur tambah hero, lihat data hero, ubah hero, hapus hero, mengurutkan hero dari nama menggunakan descending,mengurutkan level hero menggunakan ascending,mengurutkan role hero menggunakan ascending dan keluar dari program. Setiap fitur dijalankan melalui subprogram masing-masing, dan setelah selesai program kembali ke menu utama hingga pengguna memilih keluar sehingga program berakhir.



2. Deskripsi Singkat Program

Program ini dibuat untuk membantu pengguna dalam mengelola data hero Mobile Legends secara sederhana menggunakan bahasa pemrograman C++. Program ini menyediakan fitur CRUD (Create, Read, Update, Delete) yang memungkinkan pengguna untuk menambahkan data hero baru, melihat daftar hero dalam bentuk tabel, mengubah informasi hero, dan menghapus data hero yang sudah tidak meta. Selain itu, program ini juga bisa mengurutkan hero berdasarkan nama,level,role hero menggunakan ascending atau descending dan juga program ini dilengkapi dengan fitur login untuk membatasi akses pengguna sehingga hanya pengguna yang memiliki username dan NIM yang benar yang dapat menggunakan program. Dengan adanya program ini, pengelolaan data hero menjadi lebih terstruktur, mudah dibaca, dan lebih rapi.

3. Source Code

```
#include <iostream>

#include <iomanip>

using namespace std;


struct Hero{

    string nama;

    string role;

    string kesulitan;

    int level;

};


Hero hero[100] = {

    {"Ling","Assassin","Hard",90},

    {"Tigreal","Tank","Easy",70},

    {"Kagura","Mage","Hard",85},

    {"Beatrix","Marksman","Medium",80},
```

```
    {"Fredrinn","Fighter","Medium",75}
};

int jumlahHero = 5;

bool login(){
    string username, password;
    int percobaan = 0;

    while(percobaan < 3){
        cout<<"=== LOGIN ==="<<endl;
        cout<<"Masukkan Nama : "; cin>>username;
        cout<<"Masukkan NIM  : "; cin>>password;

        if(username == "050" && password == "050"){
            cout<<"Login berhasil!"<<endl;
            return true;
        }

        percobaan++;
        cout<<"Login gagal! Sisa percobaan "<<3-percobaan<<endl;
    }
    return false;
}
```

```
void garis(){
    cout<<"+----+---------------+---------------+-----------+--------+"<<endl;
}

void tambahHero(Hero *hero, int *jumlahHero){
    cout<<endl<<"=== TAMBAH HERO ==="<<endl;

    cout<<"Nama Hero : "; cin>>hero[*jumlahHero].nama;
    cout<<"Role Hero : "; cin>>hero[*jumlahHero].role;
    cout<<"Kesulitan : "; cin>>hero[*jumlahHero].kesulitan;
    cout<<"Level     : "; cin>>hero[*jumlahHero].level;

    (*jumlahHero)++;
}

void lihatHero(Hero *hero, int jumlahHero){
    cout<<endl<<"=== DATA HERO ==="<<endl;

    garis();
    cout<<"| No | Nama Hero     | Role          | Kesulitan | Level |"<<endl;
    garis();

    for(int i=0;i<jumlahHero;i++){
```

```
        cout<<"| "<<setw(2)<<i+1<<" "
            <<"| "<<setw(13)<<left<<hero[i].nama
            <<"| "<<setw(13)<<left<<hero[i].role
            <<"| "<<setw(9)<<left<<hero[i].kesulitan
            <<"| "<<setw(5)<<hero[i].level<<" |"<<endl;
    }

    garis();
}

void ubahHero(Hero *hero, int jumlahHero){
    int nomorhero;

    lihatHero(hero, jumlahHero);

    cout<<"Pilih nomor hero : ";
    cin>>nomorhero;
    nomorhero--;

    cout<<"Nama baru : "; cin>>hero[nomorhero].nama;
    cout<<"Role baru : "; cin>>hero[nomorhero].role;
    cout<<"Kesulitan baru : "; cin>>hero[nomorhero].kesulitan;
    cout<<"Level baru : "; cin>>hero[nomorhero].level;
}
```

```
void hapusHero(Hero *hero, int *jumlahHero){
    int no;

    lihatHero(hero, *jumlahHero);

    cout<<"Pilih nomor hero : ";
    cin>>no;
    no--;

    for(int i=no;i<*jumlahHero-1;i++){
        hero[i] = hero[i+1];
    }

    (*jumlahHero)--;
}

void sortingNamaDesc(Hero *hero, int jumlahHero){
    for(int i=0;i<jumlahHero-1;i++){
        for(int j=i+1;j<jumlahHero;j++){
            if(hero[i].nama < hero[j].nama){
                swap(hero[i], hero[j]);
            }
        }
```

```
    }
    cout<<"Sorting Nama Descending berhasil!"<<endl;
}

void sortingLevelAsc(Hero *hero, int jumlahHero){
    for(int i=0;i<jumlahHero-1;i++){
        for(int j=i+1;j<jumlahHero;j++){
            if(hero[i].level > hero[j].level){
                swap(hero[i], hero[j]);
            }
        }
    }
    cout<<"Sorting Level Ascending berhasil!"<<endl;
}

void sortingRoleAsc(Hero *hero, int jumlahHero){
    for(int i=0;i<jumlahHero-1;i++){
        for(int j=i+1;j<jumlahHero;j++){
            if(hero[i].role > hero[j].role){
                swap(hero[i], hero[j]);
            }
        }
    }
    cout<<"Sorting Role Ascending berhasil!"<<endl;
```

```
}


int main(){


    if(login()==false){
        cout<<"Kesempatan login habis"<<endl;
        return 0;
    }


    int pilihanmenu;


    do{
        cout<<endl<<"=== MENU HERO MOBILE LEGENDS ==="<<endl;
        cout<<"1. Tambah Hero"<<endl;
        cout<<"2. Lihat Hero"<<endl;
        cout<<"3. Ubah Hero"<<endl;
        cout<<"4. Hapus Hero"<<endl;
        cout<<"5. Sorting Nama (Desc)"<<endl;
        cout<<"6. Sorting Level (Asc)"<<endl;
        cout<<"7. Sorting Role (Asc)"<<endl;
        cout<<"8. Keluar"<<endl;


        cout<<"Pilih : ";
        cin>>pilihanmenu;
```

```
        switch(pilihanmenu){
            case 1: tambahHero(hero, &jumlahHero); break;
            case 2: lihatHero(hero, jumlahHero); break;
            case 3: ubahHero(hero, jumlahHero); break;
            case 4: hapusHero(hero, &jumlahHero); break;
            case 5: sortingNamaDesc(hero, jumlahHero); break;
            case 6: sortingLevelAsc(hero, jumlahHero); break;
            case 7: sortingRoleAsc(hero, jumlahHero); break;
            case 8: cout<<"Program selesai"<<endl; break;
        }

    }while(pilihanmenu != 8);

    return 0;
}
```

4. Hasil Output

```
=== LOGIN ===
Masukkan Nama : 050
Masukkan NIM  : 050
Login berhasil!

=== MENU HERO MOBILE LEGENDS ===
1. Tambah Hero
2. Lihat Hero
3. Ubah Hero
4. Hapus Hero
5. Keluar
Pilih :
```

```
=== MENU HERO MOBILE LEGENDS ===
1. Tambah Hero
2. Lihat Hero
3. Ubah Hero
4. Hapus Hero
5. Keluar
Pilih : 1

=== TAMBAH HERO ===
Nama Hero : zetian
Role Hero : mage
Tingkat Kesulitan (Easy/Medium/Hard): medium
```

```
=== MENU HERO MOBILE LEGENDS ===
1. Tambah Hero
2. Lihat Hero
3. Ubah Hero
4. Hapus Hero
5. Keluar
Pilih : 2

=== DATA HERO ===
+----+---------------+---------------+-----------+
| No | Nama Hero     | Role          | Kesulitan |
+----+---------------+---------------+-----------+
|  1 | Ling          | Assassin      | Hard      |
| 2  | Tigreal       | Tank          | Easy      |
| 3  | Kagura        | Mage          | Hard      |
| 4  | Beatrix       | Marksman      | Medium    |
| 5  | Fredrinn      | Fighter       | Medium    |
| 6  | zetian        | mage          | medium    |
+----+---------------+---------------+-----------+
```

```
=== DATA HERO ===
+----+---------------+---------------+-----------+
| No | Nama Hero     | Role          | Kesulitan |
+----+---------------+---------------+-----------+
| 1  | Ling          | Assassin      | Hard      |
| 2  | Tigreal       | Tank          | Easy      |
| 3  | Kagura        | Mage          | Hard      |
| 4  | Beatrix       | Marksman      | Medium    |
| 5  | Fredrinn      | Fighter       | Medium    |
| 6  | zetian        | mage          | medium    |
+----+---------------+---------------+-----------+
Pilih nomor hero yang diubah : 1
Nama baru : lancelot
Role baru : assasin
Kesulitan baru : hard
```

```
=== DATA HERO ===
+----+---------------+---------------+-----------+
| No | Nama Hero     | Role          | Kesulitan |
+----+---------------+---------------+-----------+
| 1  | lancelot      | assasin       | hard      |
| 2  | Tigreal       | Tank          | Easy      |
| 3  | Kagura        | Mage          | Hard      |
| 4  | Beatrix       | Marksman      | Medium    |
| 5  | Fredrinn      | Fighter       | Medium    |
| 6  | zetian        | mage          | medium    |
+----+---------------+---------------+-----------+
```

```
=== MENU HERO MOBILE LEGENDS ===
1. Tambah Hero
2. Lihat Hero
3. Ubah Hero
4. Hapus Hero
5. Keluar
Pilih : 4

=== DATA HERO ===
+----+---------------+---------------+-----------+
| No | Nama Hero     | Role          | Kesulitan |
+----+---------------+---------------+-----------+
|  1 | Ling          | Assassin      | Hard      |
| 2  | Tigreal       | Tank          | Easy      |
| 3  | Kagura        | Mage          | Hard      |
| 4  | Beatrix       | Marksman      | Medium    |
| 5  | Fredrinn      | Fighter       | Medium    |
+----+---------------+---------------+-----------+
Pilih nomor hero yang dihapus : 1
```

```
=== MENU HERO MOBILE LEGENDS ===
1. Tambah Hero
2. Lihat Hero
3. Ubah Hero
4. Hapus Hero
5. Keluar
Pilih : 2

=== DATA HERO ===
+----+---------------+---------------+-----------+
| No | Nama Hero     | Role          | Kesulitan |
+----+---------------+---------------+-----------+
| 1  | Tigreal       | Tank          | Easy      |
| 2  | Kagura        | Mage          | Hard      |
| 3  | Beatrix       | Marksman      | Medium    |
| 4  | Fredrinn      | Fighter       | Medium    |
+----+---------------+---------------+-----------+
```

```
=== LOGIN ===
Masukkan Nama : kugghu
Masukkan NIM  : faug
Login gagal! Sisa percobaan 2
=== LOGIN ===
Masukkan Nama : wowo
Masukkan NIM  : nyawit
Login gagal! Sisa percobaan 1
=== LOGIN ===
Masukkan Nama : keren
Masukkan NIM  : banget
Login gagal! Sisa percobaan 0
Kesempatan login habis
PS C:\PRAKTIKUM-APL\post test\post-test-apl-2>
```

```
=== MENU HERO MOBILE LEGENDS ===
1. Tambah Hero
2. Lihat Hero
3. Ubah Hero
4. Hapus Hero
5. Sorting Nama (Desc)
6. Sorting Level (Asc)
7. Sorting Role (Asc)
8. Keluar
Pilih : 5
Sorting Nama Descending berhasil!
```

```
=== DATA HERO ===
+----+---------------+---------------+-----------+--------+
| No | Nama Hero     | Role          | Kesulitan | Level |
+----+---------------+---------------+-----------+--------+
|  1 | Tigreal       | Tank          | Easy      | 70     |
| 2  | Ling          | Assassin      | Hard      | 90     |
| 3  | Kagura        | Mage          | Hard      | 85     |
| 4  | Fredrinn      | Fighter       | Medium    | 75     |
| 5  | Beatrix       | Marksman      | Medium    | 80     |
+----+---------------+---------------+-----------+--------+
```

```
=== MENU HERO MOBILE LEGENDS ===
1. Tambah Hero
2. Lihat Hero
3. Ubah Hero
4. Hapus Hero
5. Sorting Nama (Desc)
6. Sorting Level (Asc)
7. Sorting Role (Asc)
8. Keluar
Pilih : 6
Sorting Level Ascending berhasil!
```

```
=== DATA HERO ===
+----+---------------+---------------+-----------+--------+
| No | Nama Hero     | Role          | Kesulitan | Level |
+----+---------------+---------------+-----------+--------+
| 1  | Tigreal       | Tank          | Easy      | 70     |
| 2  | Fredrinn      | Fighter       | Medium    | 75     |
| 3  | Beatrix       | Marksman      | Medium    | 80     |
| 4  | Kagura        | Mage          | Hard      | 85     |
| 5  | Ling          | Assassin      | Hard      | 90     |
+----+---------------+---------------+-----------+--------+
```

```
=== MENU HERO MOBILE LEGENDS ===
1. Tambah Hero
2. Lihat Hero
3. Ubah Hero
4. Hapus Hero
5. Sorting Nama (Desc)
6. Sorting Level (Asc)
7. Sorting Role (Asc)
8. Keluar
Pilih : 7
Sorting Role Ascending berhasil!
```

```
=== DATA HERO ===
+----+---------------+---------------+-----------+--------+
| No | Nama Hero     | Role          | Kesulitan | Level |
+----+---------------+---------------+-----------+--------+
| 1  | Ling          | Assassin      | Hard      | 90     |
| 2  | Fredrinn      | Fighter       | Medium    | 75     |
| 3  | Kagura        | Mage          | Hard      | 85     |
| 4  | Beatrix       | Marksman      | Medium    | 80     |
| 5  | Tigreal       | Tank          | Easy      | 70     |
+----+---------------+---------------+-----------+--------+
```

5. Langkah-langkah GIT

5.1 GIT Add

```
PS C:\PRAKTIKUM-APL\post test\post-test-apl-2> git add .
```

5.2 GIT Commit

```
PS C:\PRAKTIKUM-APL\post test\post-test-apl-2> git commit -m "apl pt 2"
[main c880f8c] apl pt 2
 2 files changed, 1 insertion(+), 1 deletion(-)
 rename post test/post-test-apl-2/{nyoba.exe => 2509106050-MUHAMMAD_NAUFAL_ABIMAR-PT-2.exe} (51%)
```

5.3 GIT Push

```
PS C:\PRAKTIKUM-APL\post test\post-test-apl-2> git push
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 20 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 670.18 KiB | 7.70 MiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote: This repository moved. Please use the new location:
remote:   https://github.com/Nopal07A/praktikum-apl.git
To https://github.com/Nopal07A/praktikum-_apd.git
   cc18c81..c880f8c  main -> main
```